

Redis Radar

In Redis Software

Private Preview Guide

Version: v0.3.0

Deployment model: Customer-managed, on-premises (RHEL/RPM)

Audience: Approved Private Preview participants

1. Overview

Redis Radar is a centralized control plane that provides a single pane of glass for managing Redis Enterprise fleets across customer-controlled environments.

Redis Radar helps operations, platform, and infrastructure teams:

- Understand fleet-wide health across connected Redis Enterprise clusters
- Identify cluster, node, and database issues faster
- Gain basic capacity awareness
- Navigate from fleet-level visibility into cluster, node, and database-level details
- Improve operational visibility across large Redis environments

Redis Radar is intended to complement existing monitoring and observability tools. It is not a replacement for real-time monitoring, alerting, or enterprise observability platforms.

2. Private Preview Scope

Redis Radar v0.3.0 is an early Private Preview release intended for:

- Design partner validation
- Proof-of-Concept deployments
- Evaluation of core workflows
- Feedback on product direction, usability, and operational fit

Important

Redis Radar v0.3.0 is not intended for production use.

It should be deployed only in controlled, non-production environments.

Features, APIs, and UX may change without prior notice.

No guaranteed upgrade path to future releases.

Limited security validation compared with a GA release.

3. Terms and Conditions

By clicking "Accept" or by accessing, downloading, installing, or using this Preview in any way, you acknowledge that you have read, understood, and agree to be bound by the following terms.

This Preview is provided under and governed by Section 3.2 (Previews) of the Redis Software Agreement. By accepting these terms, you confirm that you have the authority to bind your organization to the Agreement.

Specifically, by installing, accessing, or using Redis Radar v0.3.0, you acknowledge and agree that:

1. Redis Radar v0.3.0 is provided as a Private Preview release.
2. Use of this Private Preview is subject to the Redis Enterprise Software Agreement, including Section 3.2, Previews.
3. Redis Radar v0.3.0 is not generally available and is not intended for production use.
4. Redis may modify, discontinue, or substantially change functionality, behavior, APIs, installation methods, or user experience in future versions.
5. Redis does not guarantee compatibility or upgradeability from this Private Preview version to future versions.
6. Participation in this Private Preview requires active collaboration with Redis, including providing feedback within two weeks of installation.

4. Key Capabilities in v0.3.0

Fleet Visibility

Redis Radar provides a unified view of connected Redis Enterprise clusters, including fleet-level health status, healthy/warning/critical indicators, aggregated risk signals, cluster connectivity and last-seen status, and entry points for drill-down investigation.

Cluster and Topology Exploration

Users can navigate from the fleet view into specific clusters to inspect cluster metadata, health, nodes, databases, connectivity status, and relevant health indicators.

Basic Capacity Awareness

Redis Radar provides high-level capacity indicators to help identify high-utilization clusters, potential capacity pressure points, and nodes or clusters that may require attention. This version does not provide advanced capacity planning, forecasting, or real-time capacity analytics.

Active-Active and Replication Awareness

- Active-Active database indication
- Replication status at the database level
- Fleet-level aggregated replication view

5. Compatibility & Authentication

Requirement	Details
Minimum Redis Enterprise version	Redis Enterprise Software 8.0.16 or later
Required APIs	health_report, usage_report, crdbs, actions
Authentication (supported)	Local authentication, LDAP, SAML
Data freshness — health	Approximately every 5 minutes
Data freshness — usage	Snapshot-based, approximately hourly
Max clusters (Private Preview)	Approximately 100 Redis Enterprise clusters

6. RPM-Based Deployment

Redis Radar provides native RPM packages for RHEL-based environments where Docker is not permitted. Services are installed as systemd units and run directly on the host OS.

Architecture

Component	Description
redis-radar-api	Application service — provides the REST API and web UI (systemd unit: redis-radar-api)
redis-radar-worker	Background worker — handles async tasks and cluster data collection (systemd unit: redis-radar-worker)
PostgreSQL 16	Database — stores application state, configuration, and collected fleet data (provided separately)

Host Requirements

Requirement	Recommendation
Operating system	RHEL 8 or RHEL 9 (x86_64)
CPU	2 vCPU minimum
Memory	8 GiB RAM minimum
Disk	20 GiB available for application data and logs
PostgreSQL	PostgreSQL 16 (must be installed and running before Radar install)
Network	Outbound access to Redis Enterprise cluster management endpoints (port 9443 by default)

Prerequisites

Before installing Redis Radar, confirm that:

- The host runs RHEL 8 or RHEL 9 (x86_64)
- The host meets the minimum sizing requirements above
- PostgreSQL 16 is installed and a radar database and user exist
- The host can reach the Redis Enterprise cluster management endpoints
- Required ports are open between Redis Radar and connected Redis Enterprise clusters
- TLS connectivity is available to target Redis Enterprise clusters
- You have credentials with appropriate permissions for each target Redis Enterprise cluster

- Your organization has approved the use of this Private Preview in a non-production environment

Installation Package

Redis will provide the Redis Radar Private Preview RPM installation package. The package includes:

- redis-radar-api-<version>.x86_64.rpm — API and UI service
- redis-radar-worker-<version>.x86_64.rpm — background worker service
- redis-radar-migrate-<version>.x86_64.rpm — database migration runner
- Checksums, release notes, and installation instructions

Step 1 — Prepare PostgreSQL

If PostgreSQL 16 is not already installed, install it and create the Redis Radar database and user:

```
# Install PostgreSQL 16
sudo dnf install -y
https://download.postgresql.org/pub/repos/yum/reporpms/EL-$(rpm -E
%rhel)-x86_64/pgdg-redhat-repo-latest.noarch.rpm
sudo dnf -qy module disable postgresql
sudo dnf install -y postgresql16-server postgresql16

# Initialize and start
sudo /usr/pgsql-16/bin/postgresql-16-setup initdb
sudo systemctl enable --now postgresql-16

# Create database and user
sudo -u postgres psql <<EOF
CREATE USER radar WITH PASSWORD '<your-password>';
CREATE DATABASE radar OWNER radar;
GRANT ALL PRIVILEGES ON DATABASE radar TO radar;
EOF
```

Note the connection string — you will need it in Step 3:

```
postgres://radar:<your-password>@localhost:5432/radar?sslmode=disable
```

Step 2 — Install the RPM packages

Copy the RPM packages to the target host, then install them:

```
sudo rpm -ivh redis-radar-migrate-<version>.x86_64.rpm \
redis-radar-api-<version>.x86_64.rpm \
redis-radar-worker-<version>.x86_64.rpm
```

The packages install service binaries, default configuration files, and systemd unit files. Services are not started automatically.

Step 3 — Configure the environment

Edit the environment configuration file at `/etc/redis-radar/radar.env`:

```
sudo vi /etc/redis-radar/radar.env

# Required: PostgreSQL connection string
DATABASE_URL=postgres://radar:<your-password>@localhost:5432/radar?sslmode=disable

# Required: credential encryption key (exactly 32 bytes)
# Generate with: openssl rand -hex 16
CREDENTIAL_ENCRYPTION_KEY=<32-byte-hex-key>
CREDENTIAL_ENCRYPTION_KEY_VERSION=v1

# Required: application base URL
APP_BASE_URL=https://<your-host-or-ip>

# Optional: listening port (default 8080)
# PORT=8080
```

Important

Do not rotate `CREDENTIAL_ENCRYPTION_KEY` after initial setup.

Existing encrypted LDAP bind passwords and cluster credentials can only be decrypted with the key that encrypted them.

Store this key securely — loss of the key requires re-entering all cluster credentials.

Step 4 — Run the database migration

Apply database migrations before starting the application services:

```
sudo systemctl start redis-radar-migrate

# Check result
sudo systemctl status redis-radar-migrate
sudo journalctl -u redis-radar-migrate --no-pager
```

The migration service exits on completion. If it exits with status 0, migration succeeded. Review journal output before proceeding if it fails.

Step 5 — Enable and start the services

```
sudo systemctl enable --now redis-radar-api
sudo systemctl enable --now redis-radar-worker
```

Step 6 — Verify the installation

1. Check that both services are running:

```
sudo systemctl status redis-radar-api
sudo systemctl status redis-radar-worker
```

2. Check the health endpoints:

```
curl http://localhost:8080/healthz/startup # expect: 200 OK
curl http://localhost:8080/healthz/ready   # expect: 200 OK
curl http://localhost:8080/healthz/live    # expect: 200 OK
```

3. Review logs if a service fails to start:

```
sudo journalctl -u redis-radar-api -f
sudo journalctl -u redis-radar-worker -f
```

Step 7 — Access the UI and connect your first cluster

1. Open the Redis Radar UI in a browser using the configured APP_BASE_URL (default port 8080).
2. Create the initial admin user on first login.
3. Add a connection to a Redis Enterprise cluster. Redis Radar requires:
 - Cluster management endpoint (typically https://<cluster-fqdn>:9443)
 - Credentials with appropriate permissions on the target Redis Enterprise cluster
 - Network connectivity from the Redis Radar host to the cluster management endpoint
4. Verify connectivity using the cluster Last Seen status in the Redis Radar UI.

Networking

Redis Radar listens on the port configured in radar.env (default 8080). Recommended approach:

- Use nginx or HAProxy as a reverse proxy to terminate TLS and forward traffic to Redis Radar
- Restrict inbound access using firewalld or iptables
- Ensure outbound connectivity from the host to each Redis Enterprise cluster management endpoint (port 9443 by default)

Important

Do not expose Redis Radar directly to the public internet.
Use a reverse proxy to terminate TLS before exposing the UI externally.

Use network controls to restrict access to trusted hosts and networks.

Security Posture

Redis Radar RPM packages install services as a dedicated non-root system user (redis-radar). Verify that `/etc/redis-radar/` permissions restrict access to authorized users only. For this Private Preview:

- Deploy only in controlled, non-production environments
- Restrict UI access using network controls or a reverse proxy
- Avoid connecting production Redis Enterprise clusters unless explicitly approved by Redis and your internal security team

7. Getting Started

After installation, complete the following quick start steps:

1. Deploy Redis Radar using the RPM packages (Steps 1–6 above).
2. Create the initial admin user.
3. Access the Redis Radar UI.
4. Add a connection to a Redis Enterprise cluster.
5. Verify connectivity using the Last Seen status.
6. Explore the fleet overview.
7. Drill down into cluster, node, and database details.
8. Document feedback and share it with Redis within two weeks of installation.

8. Support During Private Preview

Redis Radar v0.3.0 is supported through an engineering-led Private Preview process. Support includes:

- Installation assistance
- Troubleshooting support
- Issue triage
- Feedback sessions
- Product and workflow discussions

This Private Preview does not include the same support model, SLA, or escalation process as a generally available Redis product. Please reach out to your Redis contact for any questions.

9. Known Limitations

Redis Radar v0.3.0 has the following known limitations:

- Not intended for production use
- On-premises, single-instance deployment only
- No high availability or backup and restore

- No guaranteed upgrade path
- No cloud integrations
- No license management or bulk operations
- No advanced capacity planning or real-time monitoring
- Features, APIs, and UX may change rapidly